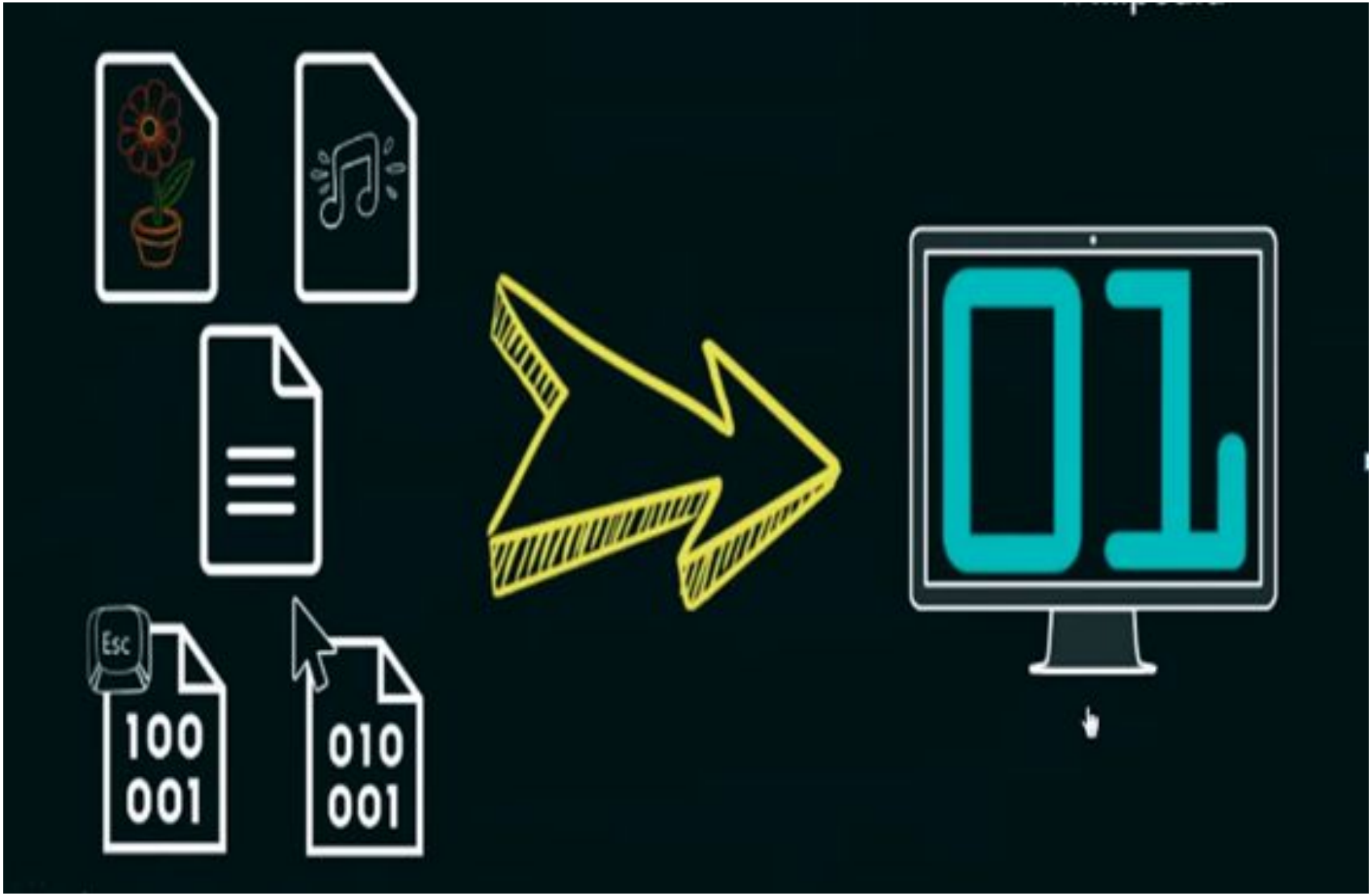
Department of Artificial Intelligence, SVNIT, Surat
Introduction to Computer science (AI101)

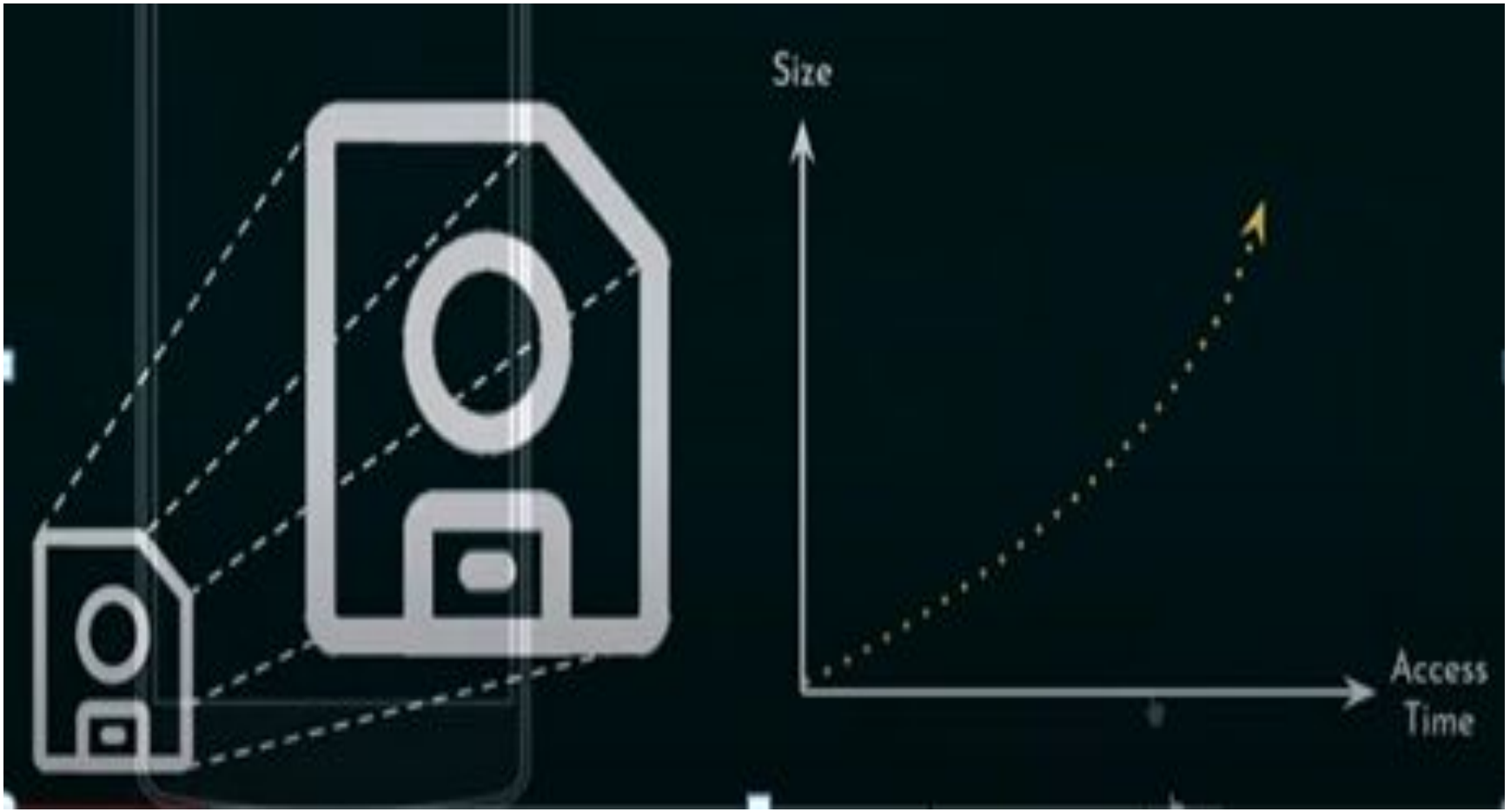
UNIT-4
MEMORY AND VARIOUS INPUT AND OUTPUT DEVICES

Introduction to Memory

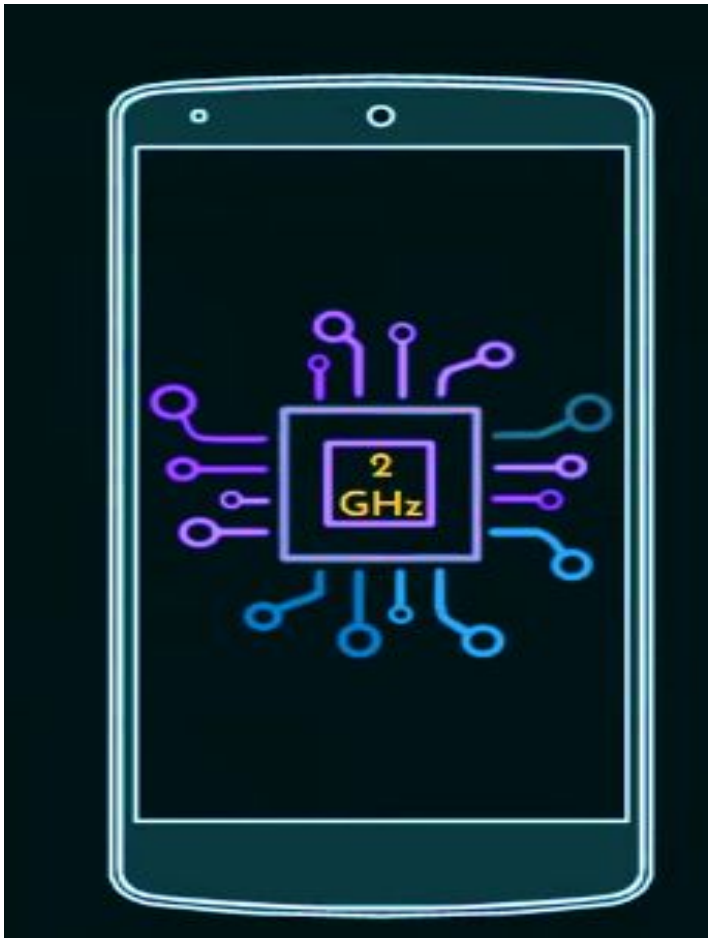
- Memory is Faculty of brain by which data or information is encoded ,stored , and retrieved when need.
- Memory devices are digital system that store data either temporarily or for a long term.


Esc
100
001
010
001
01




Size
Access
Time

Speed



- Frequency= 2GHz
- Time= 1/Frequency
  =0.5 nsec
- Processor can perform task within nsec.
- CPU is Fast.
- If memory is slower ,Most of times CPU will remain Idle. we wont have efficient machine

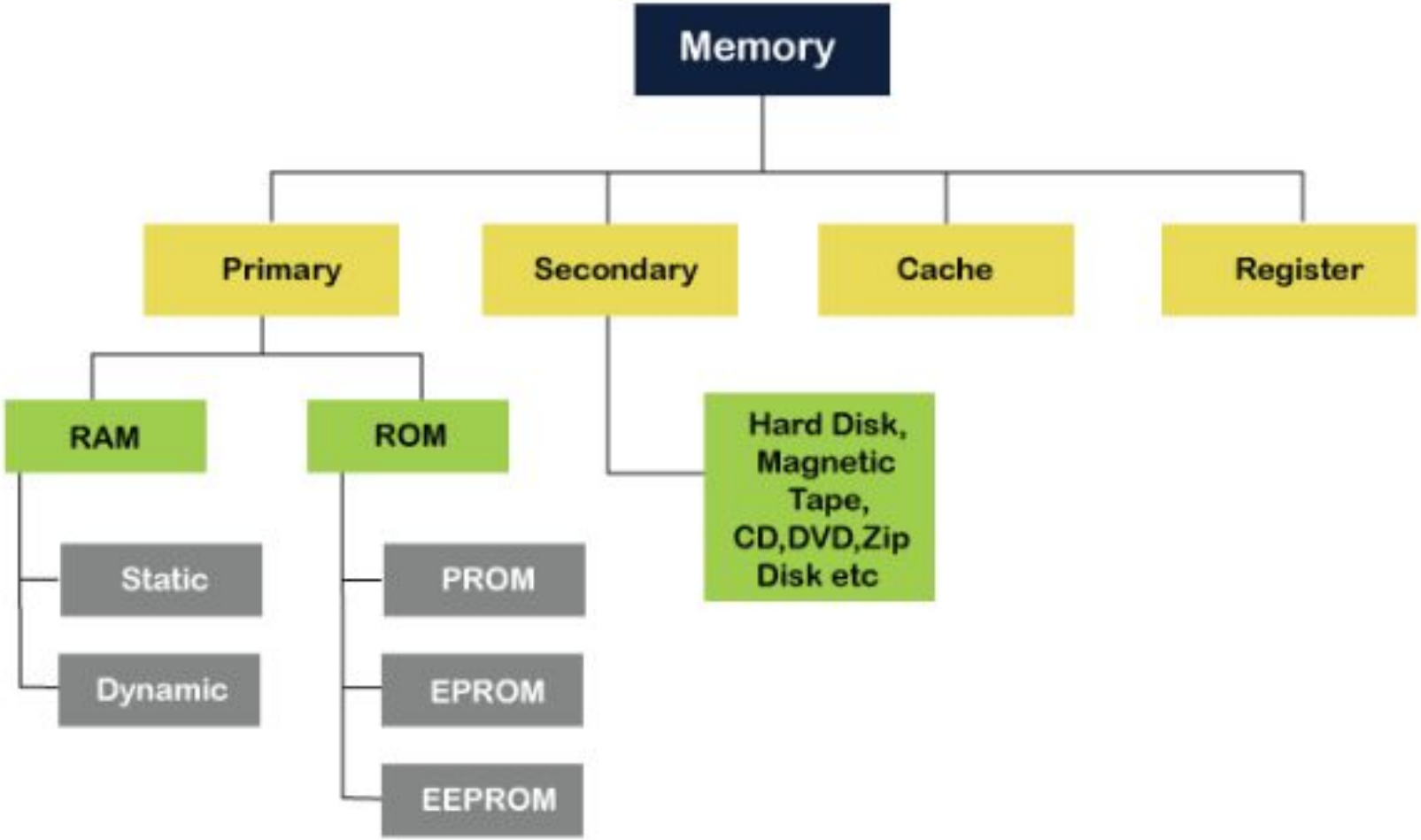
Memory Representation

- The basic unit of memory is bit. Memory can be represented in the bit and bytes..
  - 1 Bit = Binary Digit
  - 8 Bits = 1 Byte
  - 2^10 = 1024 Bytes = 1 KB (**Kilo Byte**)
  - 2^20 = 1024 KB = 1 MB (**Mega Byte**)
  - 2^30 = 1024 MB = 1 GB (**Giga Byte**)
  - 2^40 = 1024 GB = 1 TB(**Terra Byte**)

- Computer designer termed the memory to perform immediate tasks as **Primary memory**.
- The memory used more as permanent storage is known as **Secondary memory**.


Primary
Memory
OS
Secondary
Memory
SSD

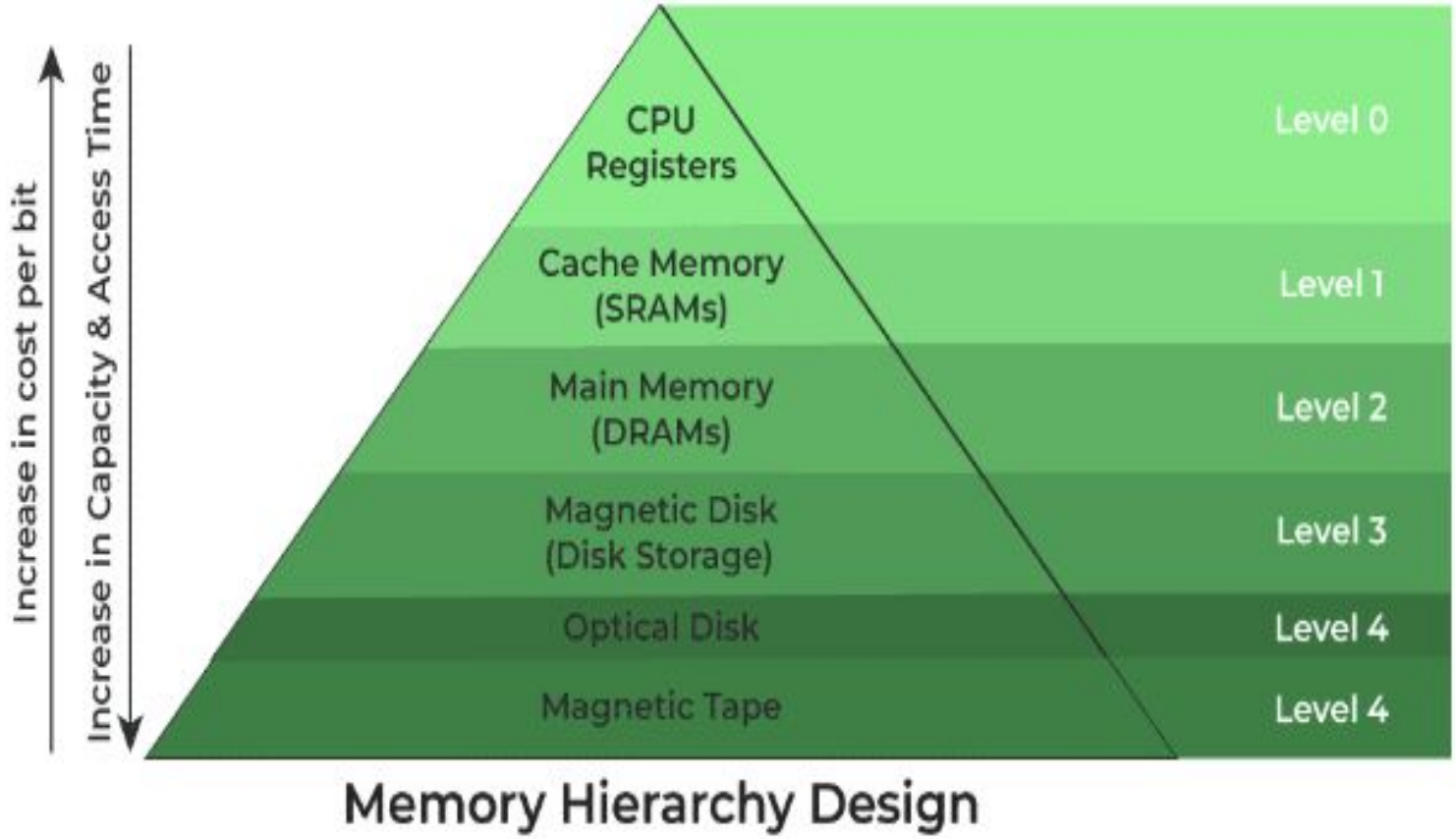
Classification of Memory



Memory Hierarchy

- In the Computer System Design, Memory Hierarchy is an enhancement to organize the memory such that it can **minimize the access time**.

- **Memory Hierarchy is divided into 2 main types as follows:**
  - Computer designer termed the memory to perform immediate tasks as **Primary memory or Internal Memory** .
  - The memory used more as permanent storage is known as **Secondary memory or External Memory**.



Memory Hierarchy Design

Storage evaluation criteria

- **Storage Capacity:**

It is the global volume of information the memory can store. It can vary from internal to external memory.

- **Access Time:**

It is the time interval between the read/write request and the availability of the data. As we move from top to bottom in the Hierarchy, the access time increases.

- **Cost Per Bit:**

As we move from bottom to top in the Hierarchy, the cost per bit increases i.e. Internal Memory is costlier than External Memory.

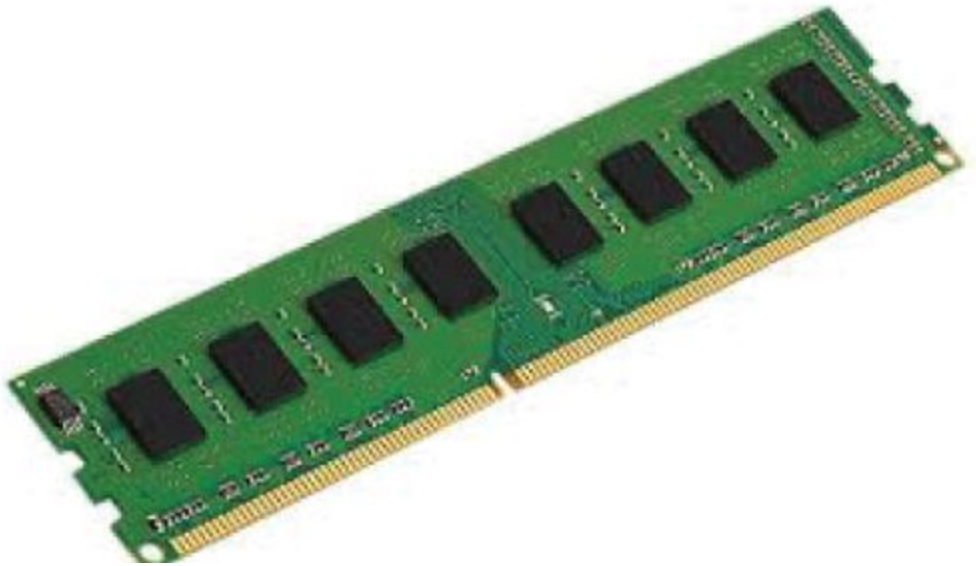
- **Physical types:** It defines the physical type of memory used in a computer such as magnetic, semiconductor, magneto-optical and optical
- **Physical characteristics:** It specifies the physical behavior of the memory like volatile, non-volatile or non-erasable memory
- **Location**
- **Access Method**

Registers

- smallest storage capacity, typically ranging from 16 to 64 bits.
- fastest access time
- Used to store the most frequently used data and instructions.

Random Access Memory (RAM)

- It is fastest types of main memory
- Volatile
- All data stored in computer memory can be read or accessed randomly at any time.



DRAM

- In DRAM, there is transistor associated with capacitor.
- Transistor can hold bit as long as its associated capacitor is charged.
- Periodic recharging is required for value retention.
- Still slow for modern processor.
- We use Cache Memory(SRAM).
- DRAM chip can hold more data than an SRAM chip of the same size.
- Both are volatile .

SRAM (Static Random-Access Memory)

- It uses flip-flops to store each bit of information.
- It does not require to refresh.
- It is faster than DRAM
- It is expensive.
- Large size
- less power consumption compared to DRAM

RAM	ROM
It is a Random-Access Memory.	It is a Read Only Memory.
Read and write operations can be performed.	Only Read operation can be performed.
Data can be lost in volatile memory when the power supply is turned off.	Data cannot be lost in non-volatile memory when the power supply is turned off.
It is a faster and expensive memory.	It is a slower and less expensive memory.
Storage data requires to be refreshed in RAM.	Storage data does not need to be refreshed in ROM.
The size of the chip is bigger than the ROM chip to store the data.	The size of the chip is smaller than the RAM chip to store the same amount of data.
Types of RAM: DRAM and SRAM	Types of ROM: MROM, PROM, EPROM, EEPROM

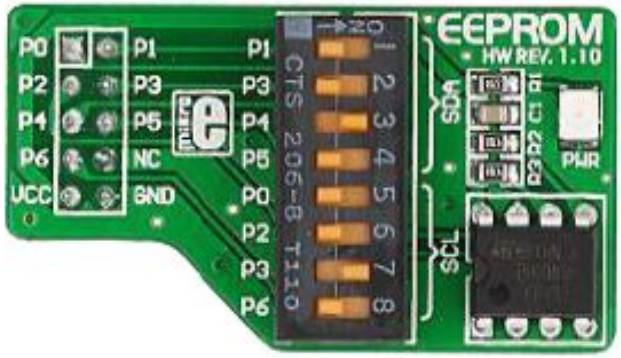
Types of Read Only Memory:

- **MROM (Masked Read Only Memory)**
  - the oldest type of read-only memory
  - program or data is pre-configured by the IC manufactures.
  - cannot be changed by the user.
- **PROM (Programmable Read Only Memory):**
  - the user can write any type of information or program only once
  - Not erasable .

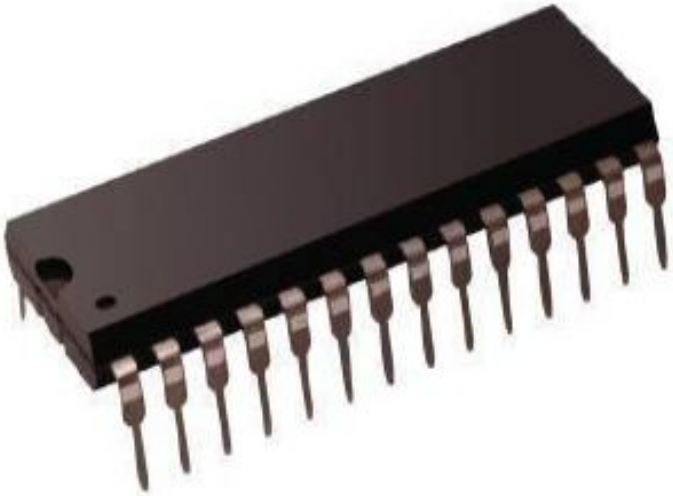
- **EPROM (Erasable and Programmable Read Only Memory):**
  - stored data can be erased and re-programmed only once in the EPROM memory by exposing the ultraviolet light for 15-20 minutes .
  - store data for a minimum of 10 to 20 years.
- **EEPROM (Electrically Erasable and Programmable Read Only Memory):**
  - Uses high voltage electrical charge
  - reprogrammed up to 10 thousand times
  - Can erase one byte at a time.
  - takes very little time to erase data, i.e., nearly 4 -10 ms(milliseconds).

Cont..

- **Flash ROM:**
  - **Non-volatile** storage memory chip.
  - Written or programmed in small units called **Block or Sector.**
  - Used to transfer data between the computer and digital devices.


EEPROM
HW REV. 1.10
P0
P1
P2
P3
P4
P5
P6
NC
VCC
GND
CTS-206-8 T110
SDA
SCL
PWR


FLASH
MEMORY
CARD
XSO-SCAN30



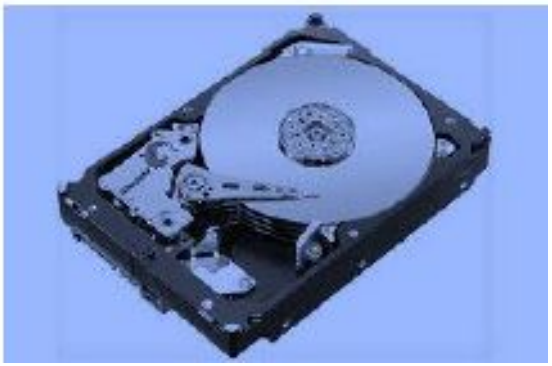

AMIC

Secondary Memory

- Known as Back up memory or Auxiliary memory.
- **Slower** than primary memory.
- Retain data **Permanently.**
- Cost –Effective.
- Bigger in size.
- Slower than Primary Memory
- Computer may run without Secondary memory.

Types of secondary storage

- **Hard Disk:**
  - Permanent storage device, Non-volatile.
  - located internally on computer's motherboard
  - Stores and retrieves data using one or more rigid fast rotating disk platters inside an air-sealed casing.



Types of secondary storage

- **Floppy Disk:**
  - consisting of thin, flexible magnetic coating disks
  - Stores electronic data such as computer files
  - only way through a new program installed on a computer or backup of the information.
  - Now a days not useful due to small storage capacity.(upto 1.44 MB)



- **CD (Compact Disc):**
  - used to store various data types like audio, videos, files, OS, Back-Up file,etc.
  - The CD has a width of 1.2 mm and 12 cm in height.
  - uses laser light to read and write data from the CDs.
  - (748 MB)



- **DVD Drive/Disc:**
  - Stands for **Digital Video Display or Digital Versatile Disc**.
  - It has the same size as a CD but can store a larger amount of data than a compact disc.(4.7 GB to 17 GB.)
  - DVD drives are divided into three types, such as DVD ROM (Read Only Memory), **DVD R** (Recordable) and **DVD RW** (Rewritable or Erasable).

- **Blu Ray Disc (BD):**
  - Optical disc storage device .
  - store a large amount of data or high definition of video recording

- **Pen Drive(**USB flash drive):



  - Portable device
  - Used to store data permanently .
  - Uses an integrated circuit chip to sto
  - Allows the users to store and transfer data
  - storing capacity of pen drives from 64 MB to 128 GB or more.


4.0
HALO

Cache Memory

- It is a small-sized chip-based computer memory that lies between the CPU and the main memory.
- faster than the main memory.
- very close to the CPU chip.
- Stores all the data and instructions that are often used by computer CPUs.

Input Devices

- An input/output device, often known as an I/O device, is any hardware that allows a human operator or other systems to interface with a computer.
- Input/output devices, as the name implies, are capable of delivering data(input) to and receiving data from a computer (output)

List of Input Devices

- Keyboard
- Mouse
- Trackball
- Joystick
- Light Pen
- Digitizer

List of Input Devices

- Scanner
- Optical Character Recognition (OCR)
- Optical Mark Reader
- Optical Bar Reader
- MICR (Magnetic Ink Character Recognition)

Keyboard


Function Keys
NUMERIC PAD
MODIFIER KEY
AIPHANUMERIC KEY
CURSOR MOVEMENT KEY

How Keyboard Works

- Keyboard is a input device used for entering data into computer by pressing one key at one time.
- It has its own processor and circuitry that carries information to and from that processor.
- The key matrix is a grid of circuits underneath



Microprocessor & Controller Circuitry of Keyboard

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- When you press a key, it presses a **switch**, completing the circuit and allowing a tiny amount of current to flow through.

- It tells the processor the position of each key in the matrix and what each keystroke or combination of keystrokes represents.



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How Keyboard Works

- Most keyboards have between 80 and 110 keys, including: Typing keys ,a numeric keypad, function keys and control keys.

- Using a keyboard, a person can type a document, use keystroke shortcuts, access menus, play games and perform a variety of other tasks.

- Keyboards can have different keys depending on the manufacturer, the operating system they're designed for, and whether they are attached to a desktop computer or part of a laptop.

- But for the most part, these keys, also called **keycaps**

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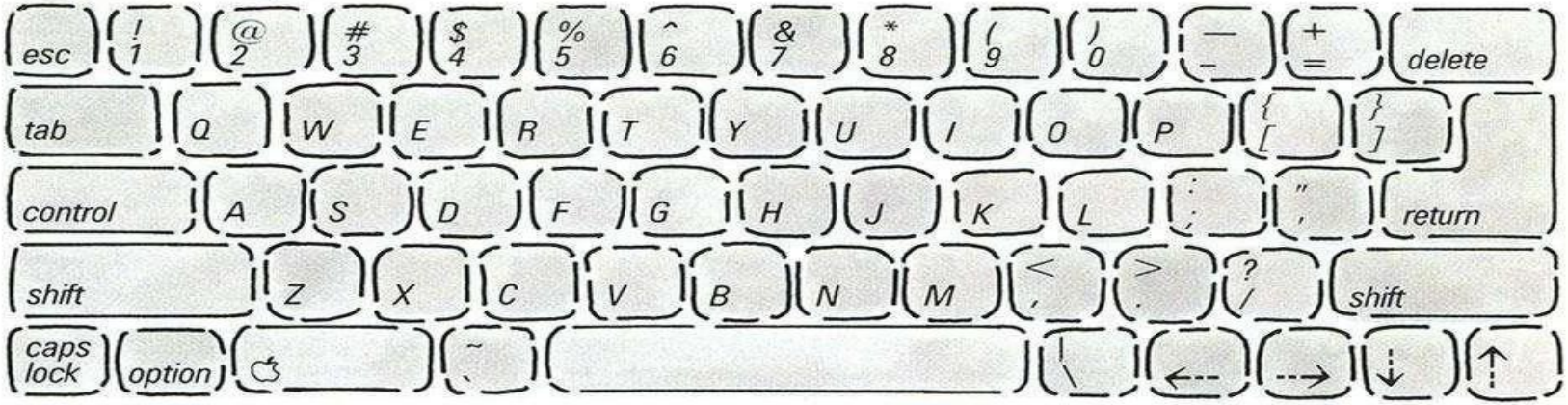
Types of Keyboard (Based on Layout)

- QWERTY Layout
- DVORAK Layout
- ABCDE
- XPeRT
- AZERTY

Each is named for the first keys in the pattern. The QWERTZ and AZERTY arrangements are commonly used in Europe.

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Standard Keyboard



Dvorak Keyboard



Source: Apple IIc Plus Owner's Guide,
Copyright 1988, Apple Inc.

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Advantages of Keyboard

- Reliable for data input of text and numbers.
- Usually supplied with a computer so no additional cost.

Disadvantages of Keyboard

- Users may be slow for not very accurate typists.
- Slow for accessing menus etc. and difficult to use if you want to move objects around the screen.
- Difficult for people unable to use keyboards through paralysis or muscular disorder.

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MOUSE

- The **mouse** is used to control the movement of a pointer on the screen when it is moved horizontally over a flat surface.

- A ball under the mouse rotates when it is moved and turns two rods, one for left/right and one for up/down.

- Buttons on the mouse let you select options from menus and drag objects around the screen etc. Some models are now wireless.

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Types of Mouse

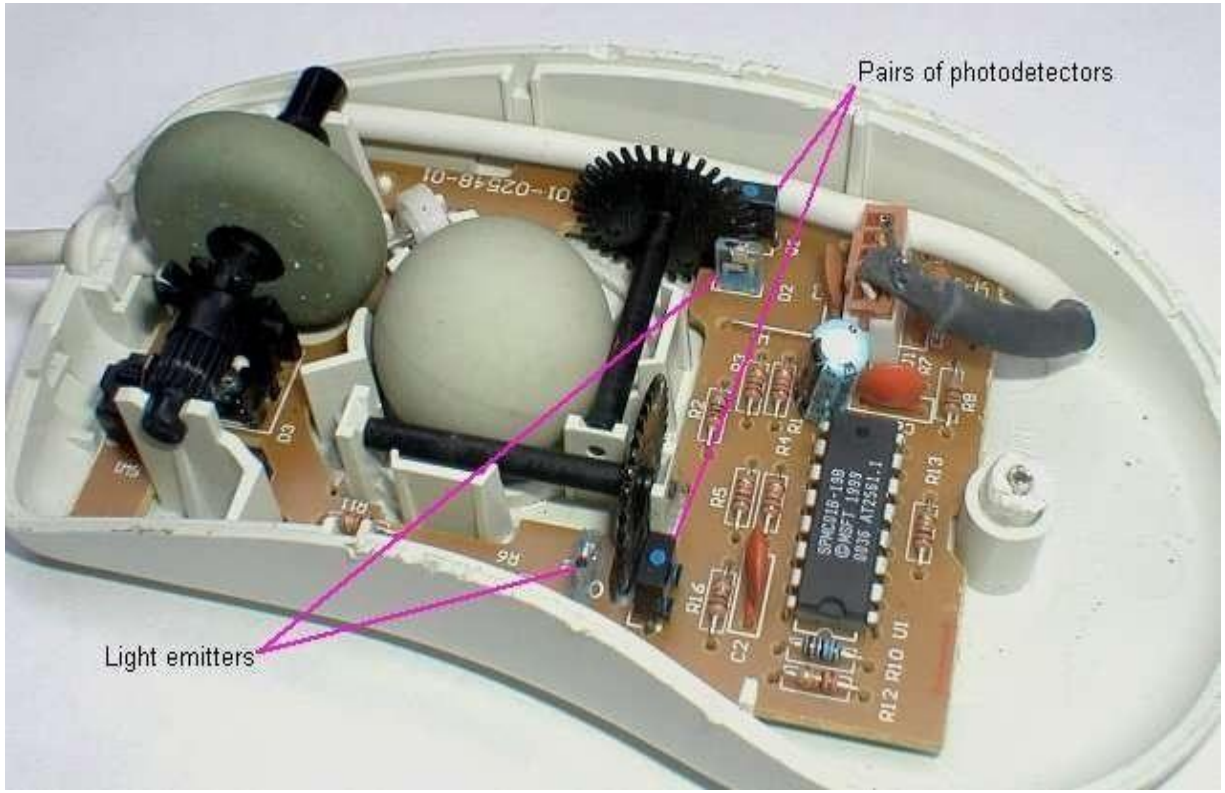
- Mechanical Mouse
  - A mouse that uses a rubber ball that makes contact with wheels inside the unit when it is rolled on a pad or desktop.
- Optical Mouse
  - A mouse that uses light to detect movement. It emits a light and senses its reflection as it is moved,

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OPTICAL MOUSE



MOUSE



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Advantages

- Moves cursor around the screen faster than using keystrokes
- Usually supplied with a computer so no additional cost.
- All computer users tend to be familiar with using them.

Disadvantages

- They need a flat space close to the computer.
- Requires moving hand from keyboard to mouse and back.

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TRACK BALL

- A **trackball** is a pointing device consisting of a ball held by a socket containing sensors to detect a rotation of the ball about two axes—like an upside-down mouse with an exposed protruding ball.

- Instead of moving the whole mouse around, the user rolls the trackball only, which is on the top or side.

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TRACK BALL





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Advantages of Trackball

- Ideal for use where flat space close to the computer is limited.
- Can be useful with laptops as they can be built into the computer keyboard or clipped on.

Disadvantages of Trackball

- Not supplied as standard so an additional cost and users have to learn how to use them.

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JOYSTICK

- A **joystick** is an input device consisting of a stick that pivots on a base and reports its angle or direction to the device it is controlling.

- Joysticks are often used to control video games, and usually have one or more push-buttons whose state can also be read by the computer.

- Joysticks are often used for playing computer games such as flight simulators. They can also be used to control the movement of a wheelchair or other machinery.

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JOYSTICK



- **Advantages of Joystick**
- Easy to learn to use. Very simple design so they can be inexpensive.

- **Disadvantages of Joystick**
- Control can be a bit crude as the directions in simple joysticks are limited to forward, backwards, left and right.
- Better models offer diagonal movement or better.

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Light pen

- A **light pen** is a light-sensitive pointing input device commonly used to select or otherwise modify text or data on a screen. Used with a CRT (cathode ray tube) monitor, these devices were an early form of manipulating and highlighting data on the screen.
- The term light pen may also refer to a **pointing input device** utilizing a light that is commonly used during a presentation. The light pen is a very focused flashlight-type of device or a laser pointer. It can direct viewers' attention to a specific area, such as a picture or text in a presentation.



Digitizer

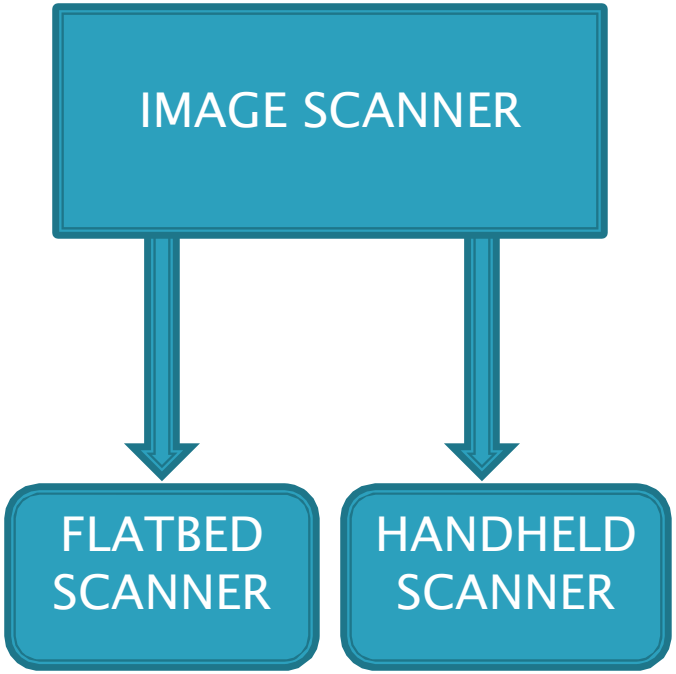
- A digitizer is a hardware device that receives analog information, such as sound or light, and records it digitally. Usually, the information is stored in a file on a computing device.

- For example, a digital camera is a digitizer. Light enters through the camera lens, and the hardware and software inside the camera converts that information to binary data, and stores it an image file. The user may then transfer the file to a computer, where he or she can edit the image, print it out, or share it online. is called digitalization.


JPG

Scanning Devices

- Scanning devices are input devices used for direct data entry in to a computer system.



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Flatbed Scanner

- A flatbed scanner is a copier machine consist of a box having a glass plate on the top and lid that cover the glass plate. For instance, a **flatbed scanner** is a peripheral device that takes an image of a document or photograph.



Handheld Scanner

- A hand held scanner has a set of light emitting diode encased in small case that can be held in hand.
  A **barcode scanner** captures binary data by progressively scanning a laser across a printed barcode.



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Touch Screen

- A **Touch screen** is an electronic visual display that can detect the presence and location of a touch within the display area.
- The term generally refers to touching the display of the device with a finger or hand.
- Touch Screens can also sense other passive objects, such as a stylus. However, if the object sensed is active, as with a light pen, the term touch screen is generally not applicable.

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Touch Screen



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Optical character recognition

- Optical character recognition (OCR) is sometimes referred to as text recognition.
- An OCR program extracts data from scanned documents, camera images and image-only pdfs.
- OCR software singles out letters on the image, puts them into words and then puts the words into sentences, thus enabling access to and editing of the original content. It also eliminates the need for manual data entry.

OCR…

- Convert physical, printed documents into machine-readable text.
- OCR systems use a combination of hardware and software.
  - **Hardware** : Copies or reads text.
  - **Software:** typically handles the advanced processing.

Optical Mark Reader

- Gathering information from human beings by recognizing marks on a document.
- It is accomplished using a hardware device (scanner) that detects a reflection or limited light transmittance on or through a piece of paper.
- OMR allows the processing of hundreds or thousands of documents per hour.

OMR

- For example, students may recall taking tests or surveys where they filled in bubbles on paper (shown right) with a pencil. Once the form had been completed, a teacher or teacher's assistant would feed the cards into a system that grades or gathers information from them.



Bar Code Reader

- **Bar code readers can read bar codes ,patterns of printed bars.**

- Lines of different widths and sizes representing data that, when read, will determine what the scanned object is. Bar codes are often used to help organize and be able to index information or prices about an object.

- A **barcode reader** or **scanner** also known as a **point of sale (POS) scanner** are hardware devices capable of reading a barcode and printing out the details of the product and/or logging that product into a database so a company can easily keep track of its inventory

Bar Code Reader

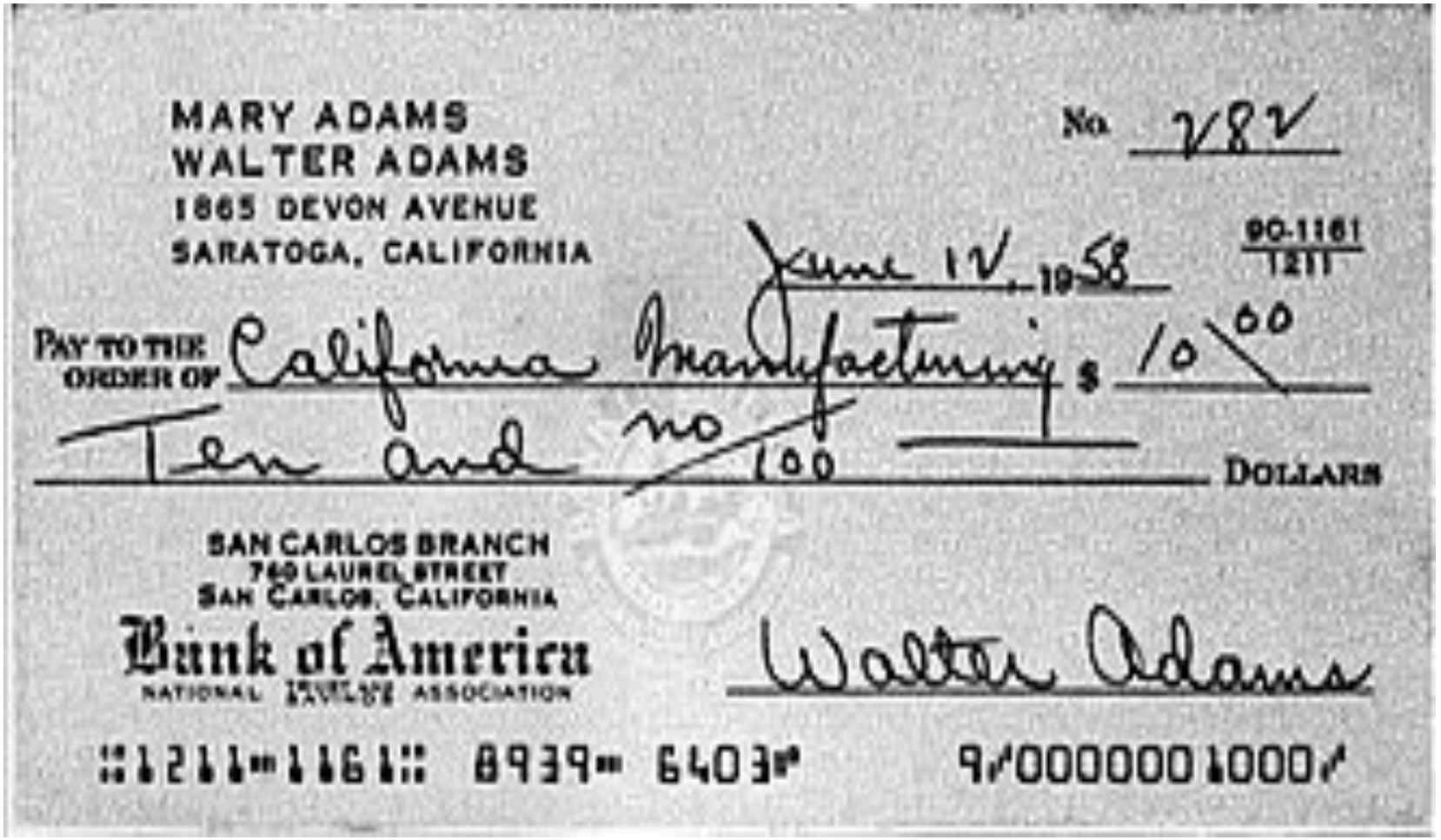
- A barcode reader consists of a scanner , a decoder , & cable used to connect the reader with a computer.
- Data coded in the form of small line are known as bar code.
- It directs a beam of light across the bar code & measures the amount of light that is reflected back.
- The scanner converts the light energy into electrical energy, which is then converted into data by decoder & forwarded to a computer.




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Magnetic ink character recognition

- Magnetic ink character recognition is the string of characters at the bottom left of a personal check that includes the account, bank branch number, and check numbers.
- MICR numbers are designed to be readable by both individuals and sorting equipment.
- They can't be faked or copied, due to the use of magnetic ink and unique fonts.
- The benefits of the technology include enhanced security against fraud and mechanization of check processing.
- MICR was developed in the late 1950s and is now in use globally.



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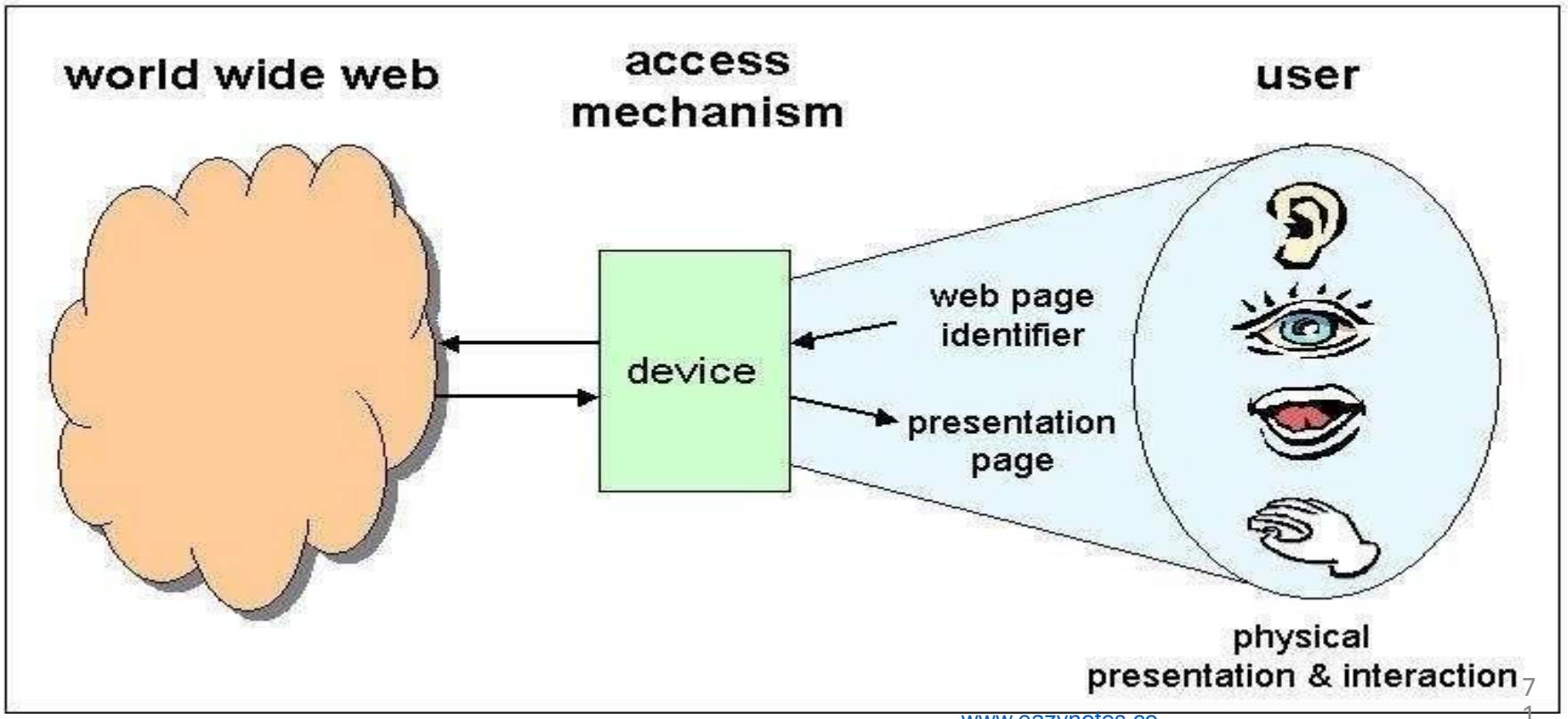
Bank of America
NATIONAL TRUST AND SAVINGS ASSOCIATION

Walter Adams

⑈1211⑈1161⑈ 8939⑈ 6403⑈ 9⑈000001000⑈

Voice Recognition Devices

- It allows a person to input data to a computer system by speaking to it.



Output Devices

List of Output Devices

- Visual Display Unit
- Printers
- Plotters
- Speakers

Output Devices

- Devices that display, print, or transmit the results of processing from the computer's memory.
- Before an output can be produced by a computer it must have an output device connected to it.

Visual Display Unit (VDU)

- A visual Display Unit (VDU) or Monitors are the most popular output devices used for producing soft-copy output.

- Monitors can display graphics, text and video. The size of a monitor is measured in inches diagonally across the screen; 15, 17, 19 and 21 inch monitors are the most common sizes. A color monitor can display from 16 to over 1 million different colors.



Visual Display Unit (VDU)

Printers

- A printer is an output device that produces a printout or hard copy of the output from a computer. There are many types of printer available.

- The speed of printers can be measured in characters per second (CPS) or pages per minute (PPM), the higher the values are, the faster the printer can produce output.

TYPES OF PRINTERS

- **Impact:**
  This type of printer works with paper via direct contact with an ink ribbon. It has similar mechanisms as of typewriter.



- **Non-Impact:**

  This Printers do not need to touch the ink ribbon to print the documents. Maintenance of Non-Impact printers is easier than Impact printers.

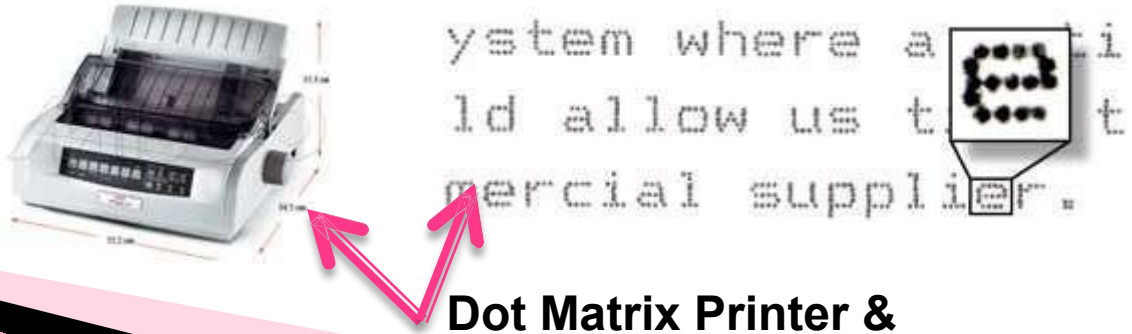


TYPES OF PRINTERS

- Daisy Wheel Printer
- Dot-Matrix Printer
- Line Printer
- Ink-Jet Printer
- Laser Printer
- Plotter

Dot Matrix Printer

- A type of computer printer with fixed number of pins or wires. this pins or wires are arranged in one or several vertical columns.
- The pins strike an ink-coated ribbon and force contact between the ribbon and the paper, so that each pin makes a small dot on the paper
- This printer is just like a typewriter. It can also create carbon copies.
- Dot matrix printers are relatively cheap .Dot-matrix printers are generally noisy.
- Slow, with speeds usually ranging between 30 to 600 characters per second.



Dot Matrix Printer & Printout

Inkjet Printer

- An inkjet printer is a peripheral that attaches wirelessly or by cable to a computer. In the home, it receives documents or images from the computer and prints on bond paper
- An inkjet printer creates an image by spraying tiny droplets of ink onto the paper. The print head moves back and forth as the paper feeds through a set of rollers. The complete image is built from many minuscule dots, like the pixels on a TV or phone screen.
- The quality of an image is determined by the number of dots per inch (DPI) the printer is capable of producing.

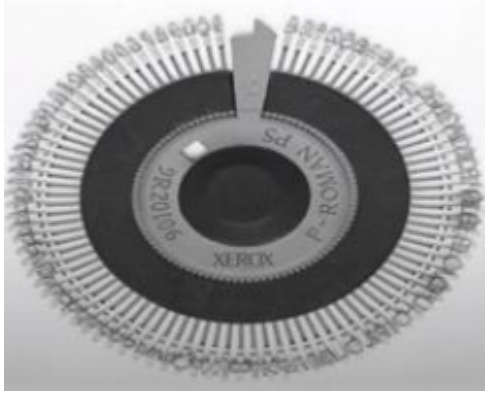


Daisy Wheel Printer

- The printer uses a metal or plastic disk containing each of the letters, numbers, and other characters it supports. When printing, the printer rotates the disk to each character and, using a hammer, strikes each character into an ink ribbon to create characters on paper.







Daisy Wheel Printer



Laser Printer

- A laser printer is a type of printer that makes use of an electrostatic digital printing process. It makes use of static electricity and toner powder in place of liquid ink. The application of toner to specific areas of the paper is determined with the help of the charge difference or static electricity.

- Laser printers can print text at speeds of four to over thirty pages per minute.

- Laser printers are quiet and fast.

Laser Printer



Plotter

- A plotter is a large printer that generates high-quality documents by moving ink pens over
  the surface of a page.

- Plotters areparticularly useful to engineers and architects, as
  they produce high-quality blueprints, maps, and floor plans.

- Can print on large printers.

- High quality printing.

- Plotters are expensive.



Plotter

Speakers

- Speakers are used to connect to a computer to generate sound, which are one of the most common output devices. Some speakers are designed to connect with any kind of sound system,.
- The primary objective of speakers is to offer audio output for the listener. The electromagnetic waves are converted into sound waves through the speaker as they are transducers.



Speakers

Thank You